SAUMYA CHATURVEDI

Computer Science Student · AI & Ml · Builder of Real-World Systems
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EDUCATION

Haridwar University 2024 – 2028
B.Tech, Computer Science — AI & Machine Learning | 2nd Year

- Relevant coursework: Data Structures & Algorithms, OOP, Computer Networks, Operating Systems

TECHNICAL SKILLS

Languages: Python, C++, Java, C, JavaScript
Libraries & Tools: OpenCV, MediaPipe, Git, VS Code,NumPy,Pandas,Matplotlib
Web: HTML, CSS
Concepts: OOP, DSA (active), Computer Vision, AI/ML, Ethical Hacking Fundamentals

PROJECTS

AlertX — Real-Time Driver Drowsiness Detection Python · OpenCV · MediaPipe · Hardware

- Built a real-time drowsiness detection system that monitors live video to flag fatigue and yawning before they become dangerous
- Implemented Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) algorithms to detect micro-level facial indicators with high reliability
- Integrated the software pipeline with physical hardware (buzzer + motor) to deliver immediate, tangible alerts — not just a UI warning
- Designed for practical deployment: low-latency processing, minimal dependencies, and straightforward hardware setup

INDEPENDENT DEVELOPMENT

Self-Directed Engineering Growth 2024 – Present

- Built from first principles — started with C, progressed through Java and OOP, then shifted to applied Python for real-world computer vision work
- Moved beyond tutorials by shipping AlertX, a functional AI + hardware system, entirely through independent research and iteration
- Currently strengthening DSA skills and exploring system design, network security, and embedded systems integration

HIGHLIGHTS

- Shipped AlertX end-to-end — from algorithm design to hardware wiring — without a course, or guided curriculum
- Gained practical expertise in real-time video processing, facial landmark detection, and sensor-actuator integration
- Proven ability to pick up new tools fast: went from zero to a working OpenCV + MediaPipe pipeline in under two months

WHAT DRIVES ME

- I build things that solve real problems — AlertX came from noticing how many road accidents are caused by drowsy driving
- Genuinely curious about how systems work at a deep level — hardware, OS, networks, not just the API surface
- Energised by hard problems: the closer a challenge is to unsolved, the more interesting it gets

CURRENT FOCUS

- Mastering DSA to competitive-programming proficiency
- Deepening work in AI systems and applied cybersecurity
- Contributing to large-scale, high-impact software — starting with a Microsoft SWE internship